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An energy-efficient cross-layer-based opportunistic routing protocol and partially informed sparse autoencoder for data transfer in wireless sensor network

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ABSTRACT

When sensor nodes in wireless sensor networks (WSNs) have limited energy, data minimization is crucial. Usually, data communications use up energy. A sensor node's lifespan can frequently be increased by sending and receiving the proper quantity of data. The PISAE (Partially Informed Sparse Autoencoder) is a state-of-the-art unmapped neural network architecture designed to reconstruct all sensor inputs from a limited number of sensors. It is presented in this publication. Use this architecture to create a system for selecting sensors. We now propose the cross-layer based opportunistic routing protocol (CORP) as an opportunistic routing mechanism for WSNs. In order to cut down on processing time and energy consumption and increase the dependability of data transfer, the best way is selected utilizing the CORP technique, which is suggested.For energy-efficient routing based combinatorial random sampling bat optimisation (ERFN-CSSBO), we also suggest using fuzzy neural networks. This enables you to conserve energy, increase the lifespan of your wireless sensor network, and keep your energy consumption in check. With CSSBO (Combined Random Sampling Prevosti Bat Optimisation), the best path is identified by integrating features (such as distance, energy, trust, and internode measurement connection stability) from all the bats.The key challenge in WSN is choosing cluster heads (CH). K-Medoid is used to enhance sensor node clustering. Important considerations include the effect on quality of service (QoS), sensor node position, proximity, and energy state needs. This study combines Hybrid BFO (Bacterial Foraging Optimization) and HSA (Harmony Search Algorithm), two well-known optimization techniques, to pick cluster heads in wireless sensor networks that are optimal in terms of distance and energy. On-task completion. The outcomes of the simulation indicate that the proposed technique improves QoS. Endpoints, throughput (1.0 Mbps), (98.5 % of packets are forwarded), and packet loss rate (1.5 %), and other QoS factors are all included in the performance statistics plotted. It performs better than conventional routing protocols in terms of network lifetime (6100 rounds), delay at both ends (1.5 s), and energy usage (30.35 mJ).

Introduction

Currently, a wide range of applications, including environmental monitoring, military surveillance, agriculture, transportation, and industry, use wireless sensor networks (WSNs) [1]. Because SN charging is not possible and the area is dangerous, some places, such deep forests or battlefields, may be challenging to enter [2]. As a result, one of the key duties of WSN to increase system longevity is to preserve SN energy. Numerous studies on the energy consumption of wireless sensor networks have been done in order to ensure the long-term operation of sensor networks [3,4]. The receiver must receive data from the WSN node. However, because it uses a lot of energy, collecting all the data from a single receiver in large-scale WSNs is challenging [5].

The receiver is utilized in a typical WSN network to distinguish between base stations and ordinary wireless sensor nodes. Due of frequent power hookups, the receiver performs better than standard nodes [6]. The energy supplied to wireless sensor nodes is typically not compensated by the WSN process, in contrast to many other wireless devices (such as mobile phones, PDAs, laptops, etc.). All sensor nodes must thus run at extremely low power levels [7].

One of the main duties of the WSN is data collection. To collect detection data in the monitoring zone, all sensor nodes routinely engage

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in distributed and coordinated detection and transmission actions [8]. Oracle predicts that WSN will produce petabytes of data. The vast amount of data that was discovered, though, was not well controlled, particularly during the data collecting procedure [9].

In comparison to the traditional k-means clustering method, the suggested method uses less energy and is more sensitive to noise and outliers. The k-means problem is resolved by the proposed clustering technique [10]. Contrary to current methods, the optimization strategy we design in this research improves the WSNs' QoS parameters and data collection effectiveness. The main accomplishments of the recently established mission are listed below:

- Examine the wireless sensor network architecture and offer system and energy models to calculate the amount of energy needed to transmit and receive data over the network.Fuzzy neural networks integrate three characteristics of routing centers: picking a hop, to improve efficiency.
- K-medoid can be used to classify networks. The optimum option for CH is provided by this technology, which makes use of distributed energy-saving cluster technology. Nodes with similar data should be grouped using the k-medoid method. We propose utilizing the BFHSA model to select the CH for each cluster node.
- The recommended BFO-HSA technique makes use of the strong convergence and dynamic characteristics of BFO, as well as the high search efficiency and high search efficiency of HSA, to select WSN cluster heads that consume less energy.

Literature survey

Some of the most recent research issues covered in this section include simple data gathering, Routing, WSN denoising, and grouping. There are now low-power WSN routing protocols [11]. To avoid duplicate transmission of similar data, each WSN node must deliver the acquired data to the sink node. Additionally, bandwidth usage and network congestion could happen. Large wireless sensor networks frequently incorporate sensors into constrained batteries, which makes quick battery charging challenging. An effective fuzzy logic method is used to address the issue of excessive energy use. suggested a congestion control-based optimal routing scheme for wireless sensor networks [12].

The GWO (Gray Wall Optimizer) technique was created to address the CH selection issue [13]. To choose CHs based on CH balancing parameters, follow this approach. Choose your CH based on high energy after evaluating your physical fitness traits. With this technique and a variety of characteristics, homogeneous networks can be deployed. This has resulted in the addition of a load balancing element to the network. SN won't be the first node to leave the network by reducing needless energy use. In [14], developed a data collection approach based on DAE (Denoising Autoencoder) to overcome the denoising issue. Data reconstruction and measurement matrices based on previous sensing data can only be evaluated during the pre-training stage of DAE.

When selecting channels for sensor-enabled IoT environments, fuzzy logic systems are crucial. According to Gupta et al. CH node selection, fuzzy logic is used and is dependent on the node's density, centrality, and energy level [15]. This method is distinct from the conventional LEACH method. In this manner, the base station picks only CH nodes for further data analysis and uses a fuzzy inference system of the Mamdani type that produces reject outputs to find advantageous nodes. The next stages are exactly the same as those in LEACH. A new CH selection method with local distance and residual energy as parameters was put out by CHEF [16].

When employing the suggested Taylor family for multi-hop routing, GWO (Gray Wolf Optimization technique) can be utilized to calculate the ideal number of hops [17]. This system is designed for multi-hop routing, which uses intermediary nodes situated in each cluster to send data from source nodes to CHs while using less energy. When employing SVRDO-SIDNL (Shift Invariant Deep Neural Learning), [18] presented Support Vector Regression Dragonfly optimization to lower latency and enhance data transfer.

PSO (Particle Swarm Optimization)-based coverage control system has been suggested [19]. The area to which each grid applies its energy usage determines the detection radius of each SN. To extend the life of wireless sensor networks, an energy-efficient trajectory scheduling method based on the coverage of a sizable number of mobile receivers was developed [20].

Problem statement

WSN calls for the gathering of data. To help with the measurement and allocation process, we regularly gather SNs. Effective data collection is challenging, especially given the enormous amount of data identified. The weak connections and limited computing power of the WSN nodes make data collection even more challenging. Compression detection (CS), a well-liked and dependable method, lowers the error rate of wireless sensor networks by data compression and denoising. Only a portion of the original data is retrievable with CS. These little volumes of data also exhibit common or low-level properties. Applying CS directly for data interpolation, however, does not deliver high performance in terms of power consumption because WSN has its own data loss pattern. The introduction of new data recovery technologies is essential. Batteries power SNs, which collect environmental data and transmit it to WSN receivers.

Proposed method

Network model

A WSN system is made up of many base stations (BS) and SNs. WSN divides the sensors into clusters to reduce power usage. Two categories are used to categorize all SNs. The common node and the CH node are separate nodes. Ordinary nodes frequently scan the surroundings to link with CH nodes. The task of gathering and transferring the data to the base station after it has been received from the public node is assigned to a CH node, which is selected from the public nodes by consensus.

Fig. 1 depicts the proposed WSN system design. The main duty of the CH is to collect data or information from many cluster nodes and transmit it to the base station. Clustering can be used to block direct

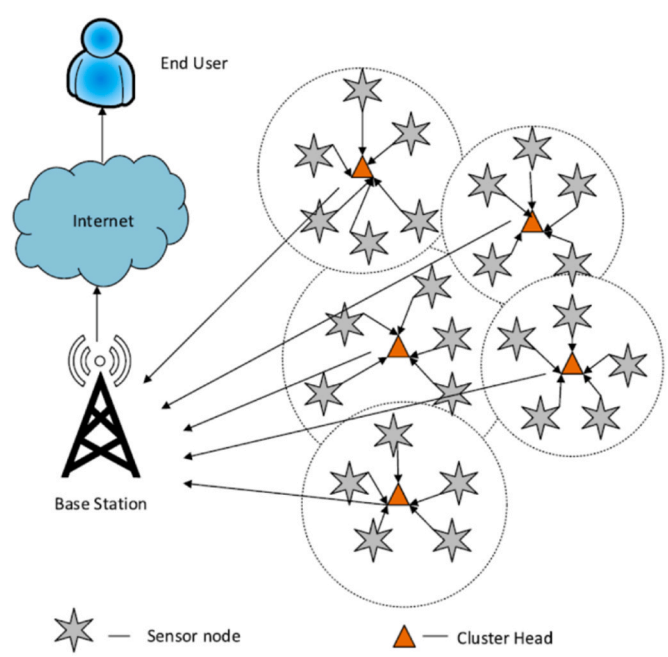

Fig. 1. WSN with cluster heads and sensor nodes.

connection between receivers and sensors.

Energy model

The amount of energy needed for several processes, including the sending, gathering, and receiving of data, is calculated. This article represents energy consumption using a straightforward radio model. The fundamental wireless model is an effective and preferable energy consumption model. Eqs. (1) and (2) below illustrate the energy needed for T-bit message transmission between the CH node and the public node.

$$E_{tx}(T,l) = E_{elec} \times T + \varepsilon_{am} \times T \tag{1}$$

$$E_{rx}(T) = E_{elec} \times T \tag{2}$$

$E_{tx}(T,l)$ l) represents the energy needed to send a T-bit message to a receiver at a distance of 1 m, while $E_{rx}(T)$) represents the energy needed to receive a T-bit message. The power needed to execute the circuit in the transceiver is stated in E_{elec}, whereas the transmit amplifier uses amps to measure its power consumption. The letter Am stands in for the amplifier's energy. To calculate the amplifier energy during the gearbox stage, utilise Eq. (3).

$$\varepsilon_{Am} \leftarrow \begin{cases} \varepsilon_f \times l^2, when & l \le l_0 \\ \varepsilon_p \times l^4, when & l > l_0 \end{cases} \tag{3}$$

Here, p and f, respectively, stand in for the multipath fading and free space model amplification coefficients. The letter lo stands in for the SN transmission model threshold. A measurement of the threshold lo is:

$$l_0 = \sqrt{\frac{\varepsilon_f}{\varepsilon_p}} \tag{4}$$

$E_{initialize}$= 2x establishes an SN's initial energy, and k-medoid employing the AHHO algorithm is advised to reduce energy consumption and extend the lifespan of WSNs.

Energy conservation using PISAE

The hierarchical network in Fig. 2 has k clusters, each of which has a cluster leader and n sensor nodes. The header may sink over great distances, preventing transmission. We suppose that clustered sensor nodes cover a large geographic region known as a sensor field while communicating only over short distances. The receiver receives the aggregated detection data from the cluster head for additional examination and/or compression. Receivers that process the data send the data to distant stations once it has been acquired from numerous cluster heads. The suggested design assumes that the receivers, cluster leaders, and sensor nodes of the sensor network are static.

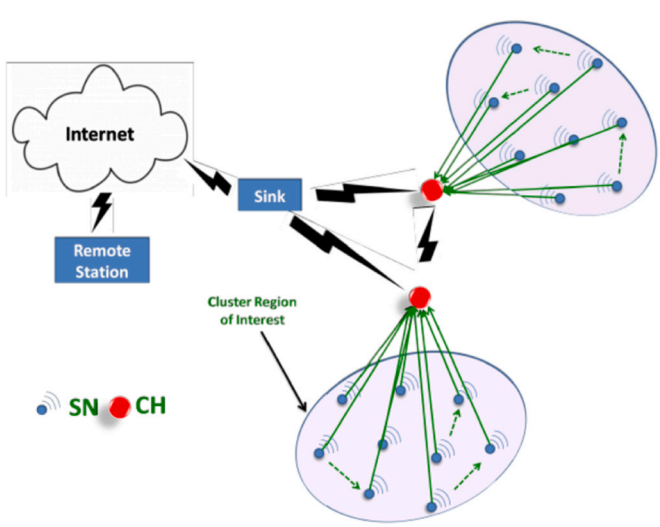

Fig. 2. Sensor network architecture.

PISAE, which was trained using the suggested energy-efficient strategy that uses few clustering, sink, and encoding weights, is used to encode a portion of the input data. At the distant station, decoding weights are used to decode each broadcast. It is necessary to use a database of previously discovered data in order to train PISAE parameters properly. Since the PISAE learning model must generalize to previously missing data points, a sizable amount of training data is often required to improve the model's generalization capabilities.A recent construction that is challenging to obtain in sufficient numbers.Small- and medium-sized generalization questions should be avoided.The restriction's nature is appropriate and required.

PISAE can learn essential spatial correlations and redundancy patterns among sensors thanks to the relatively dense array of sensor nodes and the spatial correlation of readings.PISAE models built using historical data can not generalize to new inputs when underlying processes change and affect the spatial correlation of sensor nodes. To solve this problem, we need to collect new information and develop offline models. As was already said, the amount of data required for training mainly depends on the phenomenon being identified. In general, if the sensor correlation pattern is complex, more data samples are needed.

To do this objective, network designers must be able to gather enough cluster-based training data. One of the components of the sensor node that uses a lot of energy is the wireless transceiver, as was already discussed. Therefore, effective energy management is essential for long product life and efficient energy use. Therefore, trained PISAEs are responsible for designing energy-saving schemes. Each sensor node (child node) in the cluster employs the binary label of the cluster head for downstream communication. Until the sensor node receives a fresh label from the cluster head, it will continue to function with label 1. In this case, the sensor node enters power-saving mode and turns off all energy-intensive circuits. The cluster head concurrently resets all measurements from non-critical sensor nodes to 0 for all sensors. For upstream communication, each and every cluster head employs fW1 encoding. Each member's data is encoded, compressed, and sent to the target through b1g. The cluster head also transmits two other pieces of information, including whether the data was sent on behalf of a different cluster head than the source.

Online uncommon code encoding and decoding is made possible by the temporal complexity of sensor selection and feature encoding for a single cluster head while decoding distant data, which is O(n + n′n). O (n′n) symbolises the site's complexity.Calculate Life Extension Factor (LEF) using the same evaluation formula.

$$LEF = \frac{Total \quad number \quad of \quad sensors}{Number \quad of \quad sensor \quad used \quad for \quad reconstruction} \tag{5}$$

$$LEF = \frac{n}{\sum_{i=1}^{n} (\psi_i)} \tag{6}$$

CORP to determine the best route in WSN

Due to its scalability, effectiveness in reducing packet delay, and capacity to boost communication link interdependence, CORP is employed for data transfer across the shortest path. Numerous issues exist with conventional cross-layer architecture-based routing techniques. Wireless sensor networks have a number of disadvantages, such as limited processing capability, difficulty integrating, high energy consumption, and a harder time conveying data when subjected to additional limitations.

Since CH nodes are based on residual energy data, the network layer selects nodes with high residual energy and good distribution areas. Using link quality commands, physical layer residual energy information, and link layer link quality criteria, the network system generates

routing options and selects the best routable capabilities. The network layer uses routing to locate the next-hop node in the physical layer before delivering data. The physical layer determines the available transmit power and sends this information at a reduced transmit power in order to maintain the network connection.

Regional broadcasting

ID, BS hop count, routing expense, and rare HELLO messages are some examples. If the base station path cost is set to zero and there are an infinite number of connections, each node can be spun. Eq. (7) can be used to assess the average RSSI in order to assess the connection quality metric Mn,m:

$$M_{n,m} = X_{mtp} - X_{arp} \tag{7}$$

In this instance, X_{mtp} stands for the maximum transmit power, X_{arp} for the average received signal power, and $M_{n,m}$ for the link's channel gain (n,m). We assess the link quality characteristics before estimating the corresponding link cost with the Lagrange multiplier technique.

$$\cos(n,m) = X_{cir} + X_{arp} + \frac{INI}{v} \cdot \frac{1}{h_{nm}} \tag{8}$$

where $I = -\rho 4Fn \cdot \frac{1}{InJ_x} Jx$, the expected end-to-end success probability and Fn, which stands for the cost function.

Routing method

The parent Par(n) must be selected as the following hop SN to convey information from a certain node n to that node n's BS. Node n selects a parent node after evaluating its neighbor list and determining the root BS's cost reduction in Eq. (9).

$$Par(n) = \arg_{m \in N_i(n)} \min[cost(m) + cost(n,m)] \tag{9}$$

Eq. (28) illustrates the generation of a cost-based path once the connection cost function and parent technology have been selected in this manner. After every round, a parent's choice is also updated.

Transmit power control (TPC)

The ideal transmit power is estimated in Eq. (10).

$$X_{tx(n)} = \frac{1}{h_n}[InI + In(h_n \sum_{n=1}^{H} \frac{1}{h_n})] \tag{10}$$

where H(n) is the number of hops prior to the sink. The upper bound for the distributed implementation can be obtained from the Eq. (11) below using the logical simplification.

$$(\sum_{n=1}^{H} \frac{1}{h_n}) InI \le \sum_{n=1}^{H} X_{tx(n)} \le (\sum_{n=1}^{H} \frac{1}{h_n})(InI + InH) \tag{11}$$

In order to calculate the upper bound for a given SNn, the transmit power of node n to transmit data or relay data packets to parent node m is then expressed as $X_{tx(n)}$. The following should be the $X_{tx(n)}$ Eq. (12):

$$X_{tx(n)} = \frac{1}{h_{nm}}[InI + InH(n)] \tag{12}$$

As a result, there is less packet loss or delay, and the most energy-efficient path is established for data transmission and reception.

Network and energy model

Assume N sensors are randomly placed in a network field to periodically monitor position and physical characteristics. Data is relayed to a sensor's neighbor, or one of its neighbors. Assume that the sensor has the same initial energy when it stops. Each sensor has the same amount of processing power. Think about a wireless link that symmetrically connects two proximity sensors. positioned within the recipient's network.R should represent the highest practical transmission rate for each sensor. Along with adaptive transmission, the distance between two adjacent sensors is taken into consideration. We describe a first-order radio model to evaluate the energy usage of suggested routes. The bit size of the packet is denoted by M. When the initiator sensor is d meters away from any of its surrounding sensors, the amount of energy required to transmit an m-bit packet is expressed as follows:

$$E_{TX}(m,d) \leftarrow \begin{cases} m * E_{elect} + m * \varepsilon_{fsp} * d^2, if & d < d_0, \\ m * E_{elect} + m * \varepsilon_{fsp} * d^4, if & d < d_0, \end{cases} \tag{13}$$

The following amount of energy is required to receive an m-bit packet:

$$E_{RX}(m) = m * E_{elect} \tag{14}$$

In this case, E_{Select} represents information about the energy lost during each electron bit's transit. Numerous factors, including the permitted bitrate, digital encoding, and modulation, have an impact on ESelect. The symbols for the energy requirements in free space path and multipath settings are f sp and mpf, respectively. When the distance between two adjacent sensors for which energy usage is computed is less than or equal to *lo* ($lo = \sqrt{\xi_{fsp}/\xi_{mp}}$), the radio model applies (13) otherwise (14) to calculate the energy need for transmitting the data.

Routing centric parameters

Degree distribution of sensor

Understanding how they connect to nearby sensors is the primary function of WSN sensors. Data transmission from the network via zero sensors is not possible. Depending on the sensor's rating, there is a greater chance that it may communicate data across the network. It shows resistance to link mistakes. Priority is given to neighboring sensors with less favorable conditions for choosing the next hop.Check the accuracy of the sensor distribution now. We define N sensors within the network domain. The accuracy of each sensor is determined by the sum of N - 1, an independent random variable with a binomial distribution. The chance of the link forming is given by the random variable's degree, p. The history of awarding degrees is as follows.

Initial Expected Degree. When N is replaced for N-1, there is only a little amount of error due to the magnitude of N. Let's say a sensor's predicted degree is

$$E(\alpha) \approx \text{pN} \tag{15}$$

when p is very small, the probability tends to a Poisson distribution, which is denoted by the following formula:

$$P(\alpha \approx k) \cong \frac{((N-1)p)^k}{k!} e^{-((N-1)p)} \tag{16}$$

According to this concept, there is a chance that a node has at least one degree.

$$P(\alpha \ge 1) \cong = 1 - e^{-((N-1)p)} \tag{17}$$

Finding the link probability p and balancing the sensor locations in the A-regional network field represent the next difficulty. The broadcast region of each sensor affects P. The formula AB=R2 determines each sensor's coverage area. Consequently, utilize to ascertain whether a link is likely to exist.

$$P = \frac{A_B \cap A}{A} \approx \frac{\pi R^2}{A} \tag{18}$$

when p is substituted in Eq. (18), the projected degree of a sensor—which ignores border sensors—is produced.

Forward progress

The next-hop sensor is chosen by the suggested path from the nearby

sensors in the forward search space. The transparent portion of the sensor in relation to the direction of sinking is known as the forward search space (see red region in Fig. 4).The FSS membership of the sensor's peripheral sensor nodes (nj) is subsequently verified. The position of sensor n_j is shown in Fig. 3 by the point nj (x_j, y_j), whereas the location of sensor n_i is shown by the point ni (x_i, y_i). The point s with the coordinates (x_s,y_s), (x_s,y_s), represents the sink. The relationship between the two points s(x_s, y_s) and ni(x_j, y_j) is shown by the equation.

$$a_1x + b_1y + c_1 = 0 \tag{19}$$

$$a_1 = y_i - y_s \tag{20}$$

$$b_1 = x_s - x_i \tag{21}$$

$$c_1 = y_i(x_i - x_s) + x_i(y_s - y_i) \tag{22}$$

Discover the projection of the point n_j(x_j, y_j) that the system of equations specifies.Assess the water tank's development. By projecting n_j(x_j, y_j) with a perpendicular from n_j(x_j, y_j) to the line, the intersection of the lines given in the Eq. (23) is produced. This is the vertical equation:

$$a_2x + b_1y + c_1 = 0 \tag{23}$$

$$a_2 = y_i - y_s \tag{24}$$

$$b_2 = x_s - x_i \tag{25}$$

$$c_2 = y_i(x_i - x_s) + x_i(y_s - y_i) \tag{26}$$

Determine the point n_j(x_j, y_j) 's projection P(x_j, y_j) on the line given in Eq. (27).

$$(x_p, y_p) = \left(\frac{b_1c_2 - b_2c_1}{a_1b_2 - a_2b_1}, \frac{a_2c_1 - a_1c_2}{a_1b_2 - a_2b_1}\right) \tag{27}$$

If the following inequality is satisfied, a sensor n_j belongs to the FFSi of a sensor ni.

$$n_i \in FSS_i iff \tag{28}$$

$$\sqrt{(x_i - x_j)^2 + (y_i - y_j)^2} < r^2 \tag{29}$$

and

$$\sqrt{(x_s - x_p)^2 + (y_s - y_p)^2} < \sqrt{(x_s - x_i)^2 + (y_s - y_i)^2} \tag{30}$$

where the circle's radius, r stands for the sender sensor's ni transmission range.

Residual energy

Remaining energy is defined as energy that is still present after

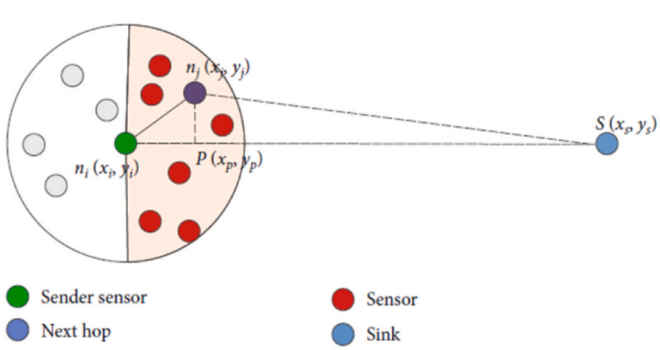

Fig. 3. Forwarding search space.

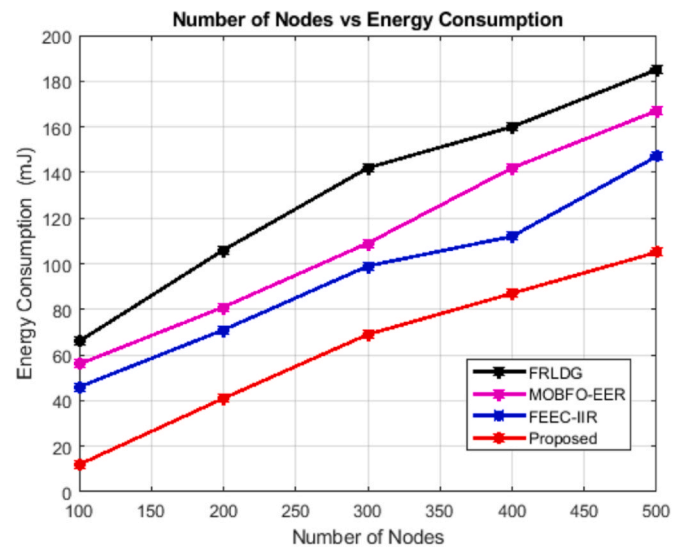

Fig. 4. Number of nodes vs energy consumption.

transfer. We advise utilising sensors powered by the most energy from the subsequent hop for this assignment. Powerful sensors have longer lengths. After receiving and sending a data packet of m-bit size, the sensor's remaining energy is denoted by the acronyms Initial Energy (E_i) and Residual Energy (E_R).

$$E_R = E_i - (E_{TX}(m, r) + E_{TR}(m)) \tag{31}$$

Expected energy consumption

When utilizing (32) to communicate m bits of data from a sensor to its subsequent hop sensor, the anticipated energy usage $E(E_{total}(m, r)$ can be written as

$$E(E_{total}(m, r) = E_{TX}(m, r) + E_{TR}(m)) \tag{32}$$

$$E(E_{total}(m, r) = \left[2e_{elect} + e_{efp}\left(\frac{n_h}{2n_h + 1}r\right)^2\right]m \tag{33}$$

ERFN-CSSBO

Finding a secure way to convey data is one of the CSSBO algorithm's steps, according to ERFN. WSNs employ broadcast transmission channels and lack a static infrastructure or central management. Wireless sensor networks are consequently extremely susceptible to assaults since they lack tamper protection. A hacker may therefore listen in on all communication, transmit harmful packets, replay earlier messages, and take over sensor nodes. Because of this, node authentication and privacy are the two key security-related challenges in applications like military communications. The novel combinatorial random sampling Prevosti bat optimisation technique is covered in this work along with security analysis of different assaults in military applications for node authentication versus security standards. Swarm intelligence is the foundation of the suggested method, which uses it to automatically locate effective solutions inside a multidimensional search field. The term "searcher" refers to a bat that begins its search for food at its present speed and position and ends it where the food is. Every bat needs to be in the bait. In this case, the bat is connected to a cluster head and given a multi-objective function. Every bat that locates and catches food is on an ideal path (i.e., location).The BAT optimisation method can efficiently identify the global optimal solution and choose the best route. Additionally, bat optimisation offers very quick convergence, which is the discovery of a stable optimal path at the conclusion of a series of solutions. This aids the algorithm in choosing the ideal route for enhancing data transmission and cutting down on latency.

A population of cluster heads (i.e., bats) named Ch1, Ch2, Ch3, and Chk are first created using the suggested approach, and they are stimulated with beginning placements and speeds across the search space (i.e., the network). Node distance, energy, dependability, and connection stability are just a few examples of the objective functions used to determine these values.

Consider a space with only two dimensions. Prevosti distance is the measurement of the separation between two nodes. A metric called prevosti distance is used to locate the closest nodes for the best path exploration.

$$D = \frac{1}{n}\sum_{i=1}^{n}\sum_{j=1}^{m}\left|sn_i - sn_j\right| \tag{34}$$

The Prevosti distance between the sensor nodes Sni and Snj is represented by "D" in this case. The number of sensor nodes is denoted by "n".

Since sensor node trust is a crucial component of identifying the most dependable nodes in many sorts of attacks, such as black holes, grey holes, floods, and scheduling attacks, sensor nodes keep an eye on their nearby neighbours and use the following confidence estimate method.

$$Tst = \left[\frac{F\left(dp_{j\to k}\right)}{S\left(dp_{i\to j}\right)}\right] \tag{35}$$

$$\varphi_F = \text{argmin}(D)\&\&\text{argmax}(E_{res}, Tst)\&\&(LC_{i\to j}(t) > T) \tag{36}$$

In (36), E_{res} is the residual energy, T_{st} is the reliability, $LC_{i\to j}(t)$ is the link stability, T is the threshold, and F is the fitness function. "Arg min" is the parameter of the minimal function. Through a random universal sampling selection procedure, the top 'n' bats are chosen.

WSN-based clustering using k-medoid and BFHSA

Clustering in WSNs is the most effective way to prolong network life and save energy. The proposed clustering algorithm is explained in more depth below.

Cluster formation using the k-medoid algorithm

Set up the clustering scheme by classifying the SNs and choosing a CH for every cluster of WSNs. Data collection and transmission to the base station from a particular cluster node are the key responsibilities of the CH. Using the k-medoid approach, one can increase communication between SNs, discover precise central clusters or cluster formation centers, and spend less energy. The procedure estimates the number of clusters and chooses the first CH node to produce the final clustering result. At first, we may determine the C number of clusters by doing the following:

$$C = \sqrt{\frac{N}{2}} \tag{37}$$

The overall number of nodes is N. The first mean point, which has been defined as follows (38), must first be assessed. Think of L as the place where all the nodes are put together.

$$L = \frac{\sum_{n=1}^{N} Z_n}{N} \tag{38}$$

Zn is used to represent sensor n's coordinate. The average separation (D) between all SNs and L can be determined using Eq. (39):

$$D = \frac{\sum_{n=1}^{N} |Z_n - L|}{N} \tag{39}$$

The centroid or center circle can be identified using D and L. At a set of randomly chosen C sites that are situated on the centroid, the first mean point Qi is measured.

$$Q_{n_x} = D \times \cos\left(\frac{360}{C} \times (n-1) \times \frac{\pi}{180}\right) \times L_x \tag{40}$$

$$Q_{n_y} = D \times \sin\left(\frac{360}{C} \times (n-1) \times \frac{\pi}{180}\right) \times L_y \tag{41}$$

In this instance, Q_{nx} and Q_{ny} stand in for the cluster n's average point coordinates. C stands for the quantity of clusters. The starting average score can be selected at random using the k-medoid method. As two or more average points get closer together, the repetition duration lengthens, producing a quadratic output. For the purpose of locating the center or center circle, the aforementioned stages are crucial since they offer a consistent first average score. The K-medoid technique uses actual nodes rather than representing CH nodes as theoretical averages. We employ partitioning strategies based on deletion of representative and widespread nodes. The absolute error criterion B is expressed in Eq. (42).

$$B = \sum_{n}^{q}\sum_{x\epsilon q_i} dis(x, v_i) \tag{42}$$

Clusters n and qi respectively serve as their representations, with x serving as the common node of cluster qi. So, the solution to this problem is greed. Let Q be the group of CH nodes, where Q is made up of Qq-1, Qq, Q1, Q2,., Qm,., and Q. The CH node qm is replaced by a public node named Qrand that is chosen at random. The energy left in this random node must be more than the total energy of all nodes. A different method of solving problem (43) is:

$$Q^{iterate+1} \leftarrow \begin{cases} Q^*, B^* - B^{iterate} < 0 \\ Q^{iterate}, otherwise \end{cases} \tag{43}$$

The network needs to be temporarily transferred to the C cluster since Q* now equals "Q1, Q2,.Qrand,., Qq-1, Qq." Eq. (36) and the adjusted population served as the basis for our evaluation of the absolute error threshold. B* is compared to the actual absolute error threshold for cluster B on each replication (iteration). It follows that B* is less than B (iterations) if Q(iterations + 1) Q* . Iterations that follow are started if the alternative is unsuccessful. To see the clustering results, repeat step 30. After the cluster building process is complete, AHHO is used to select the appropriate CH.

CH selection utilizing the harmony search algorithm and hybrid bacteria foraging optimization

Optimizing hybrid bacteria foraging

The efficiency of the search engine is what determines how effective the optimization procedure is. There are two types of searches: international and regional. Although BFO's global search tools are laborious, its local search features are efficient. In HSAs, having global search capability is crucial. As a result, the BFO and HSA combination enhances both global and local solution search. Random velocities are used by BFO during chemotaxis correction. It will take time for them to converge because it is a blind voyage. We incorporate HSA into the hybrid algorithm's recommended random rate in order to account for it. The suggested approach keeps the benefits of both HSA and BFO procedures while removing the drawbacks of BFO. HSA and LEACH are used in the hybrid BFO for the best CH selection.

A novel, physiologically based probabilistic global search technique, the BFO (Bacterial Foraging Optimization) algorithm imitates the foraging behavior of E. coli. for locating, handling, and eating food. While looking for food, bacteria may suffocate or swim. Bacteria can be made to change their course by falling. Chemotaxis is the phase of swimming in which bacteria travel in the current's direction. Natural selection mechanisms favor effective collecting procedures but reduce animals for which collection approaches are unsuccessful, hence genetic breeding is favored. Poor foraging tactics are either abandoned or

improved upon after a few generations. It is employed by researchers as an ensemble optimization method.

Harmony Search Algorithm

Harmony search algorithms have idealized the improv methods utilized by talented musicians. Musicians favor one of three methods to improve their playing: (2) Write down new or arbitrary notes. (3) Play common musical forms (including treble sequences) and memorize notes. In 2001, the three methods listed above were declared the top three mathematical algorithms. Due to the Bionic Optimizer's ability to choose the optimum match, harmonic memory usage is crucial. Your best harmonies can then serve as your current harmonies memory as a result. The Harmonic Memory Consideration Ratio (HMCR), commonly known as the accept[0,1] option, is typically specified to make greater use of this memory. The convergence will be relatively sluggish if there are few HMCRs because it will only choose a few of the best chords. Therefore, it was determined that the acceptable range was 0.70–0.95.

The ratio of pitch adjustment (RPA) and the pitch bandwidth range determine the second component, pitch adjustment. Frequency shifting is a crucial part of changing the pitch of music, even though it's relevant to creating somewhat varied HSA solutions. Randomization, the third component, increases the variety of solutions. Similar work is done through pitch adjustment, but only locally, using several pitch changes to obtain the optimal local outcome.

Hybrid BFO-HSA for cluster head selection

On BF optimization techniques and typical HSA features, the BFO-HSA CHS method is based. When using the HSA optimization technique, solutions from the locally obtained best solution are avoided in favor of PAR and HMCR solutions and metrics that can enhance the best solution that was discovered worldwide. To do this, we collaborate with BFO and HSA to address high-dimensional development and search issues, and to generate optimal solutions with respectable convergence accuracy. The suggested BFO-HSA scheme's optimal CHS flow diagram is depicted in Fig. 1.

Bacteria can move from one place to another, alter their location, and change the season after each data transfer iteration thanks to optimization employing BFO and BFO-HSA. The best solution can be determined in n regions greater speed of convergence because the dynamic nature of BFO meets the high search efficiency of HSA in each region. This functional procedure is carried out over several data transfer rounds and is BFO-HSA optimized. Each round, CH gathers all cluster members' data and transmits it to Sink for data transmission.

The process of addition has started at BFO-HSA,

Step 1: Start by employing a network strategy. replicating the original WSN's nodes. The energy of every node is completely replenished.

Step 2: Setting up the parameters Define the i-th BFO's position as a matrix assuming there are n BFOs. The matrix below (44) shows all bacterial aggregation locations.

$$BF = \begin{bmatrix} BF_{1,1} & \cdots & BF_{1,d} \\ \vdots & \ddots & \vdots \\ BF_{n,1} & \cdots & BF_{n,d}, \end{bmatrix} \tag{44}$$

where BF_(i,j) is the i-th BFO of the j-th dimension. All bacteria in the forest must have equal access to the same beginning foraging areas, as shown in Eq. (45) below.

$$BF_i = BF_{LB} + U(0,1) \times (BF_{UB} - BF_{LB}) \tag{45}$$

The upper and lower bounds of the i-th BFO are thus represented by BFOUB and BFOLB, respectively, on the jth dimension. U(0, 1), on the other hand, represents a uniformly distributed, randomly generated number that is defined in the [0,1] range. A BFO-HSA optimization technology called Bacteria Harmonic Memory (BHM) collects bacteria and initializes their harmonic memory.The original BHM is made up of the solutions to a large number of optimization problems that were produced at random. The solution to the query is the index of "k" CH. The BHM is calculated using Eq. (46). The rows of the BHM matrix represent different random answers to the current optimization issue.

$$\begin{bmatrix} BF_1^1 & \cdots & BF_k^1 \\ \vdots & \ddots & \vdots \\ BF_1^{SHM} & \cdots & BF_k^{BHM}, \end{bmatrix} \begin{pmatrix} f_{obj_1} \\ \vdots \\ f_{obj_2} \end{pmatrix} = \begin{bmatrix} B^1 \\ \vdots \\ B^{BHM} \end{bmatrix} \tag{46}$$

Step 3: Sorting, declaring, and randomly choosing: raising each BFO's set of positional fitness values. The remaining BFOs are thought to be a part of common trees. By creating the predator presence probability PDP location update mechanism, the natural behavior of the BFO was produced. Three scenarios can result in bacterial feeding activity. Each case's underlying presumptions are:

- BFOs efficiently locate their favorite prey by gliding across woodlands devoid of predators.
- Take precautions since predators exist. BFO is forced to embark on a brief, aimless walk in order to search for nearby hiding places.

Step 4: To improve the BHM after defining it in accordance with Eq. (47), build new harmonic vectors [BF'_1,BF'_2,.,BF'_k]. Each component of the newly defined harmonic matrix denoted by "Ij" is built using the defined HMCR values in Eq. (47).

$$BF_j^{'} \leftarrow \begin{cases} BF_j^{'} \in BHM \quad with \quad probability \quad HMCR \\ BF_j^{'} \in BHM \quad with \quad probability(1 - HMCR) \end{cases} \tag{47}$$

After the BHM generates BF_jˆ', BF_j may then be changed further for the PAR value or in other ways. This opens the door to an alternative strategy to the modified BHM candidate (1-PAR). The pitch of the specified BF_jˆ' changes to:

$$BF_j \leftarrow \begin{cases} BF_j^{'} \in BHM \quad with \quad probability \quad PAR \\ BF_j^{'} \in BHM \quad with \quad probability(1 - PAR) \end{cases} \tag{48}$$

The BFOnj sign in this instance denotes the closest node with an energy greater than the CH energy of each cluster.

Step 5: The dominating worst harmony is removed from the BHM pitch and the updated harmony is added to the SHM if the updated blended harmony vector's fitness value is higher than the fitness value determined from the worst harmony in the BHM.

Step 6: Sliding equilibrium, where the drag (D) and lift (L) forces are added to produce a resultant force (R) with the same magnitude and direction as the starting force, describes the sliding behavior of the bacterial bait. Bacteria can glide at a constant velocity (V) while using R to generate a linear gliding route. The optimization approach for this study was created using a crude prototype of the gliding behavior. Eq. (49) below describes the lift or sliding force and rate at which bacteria locate food at a fixed angle and constant vertical downhill velocity.

$$\frac{L}{D} = \frac{1}{\tan\phi} \tag{49}$$

BF can improve the glide rate by extending the flight path by offering a minimum glide angle (ϕ). In this experiment, lift caused by the bacterial wings causes air to be pushed downward from above. Eq. (50) has the following form:

$$L = \frac{1}{2\rho C_L C^2 A} \tag{50}$$

Here, A(154 cm2) denotes the surface area of the squirrel's body, (1.204kgm3) is the density of the air, CL denotes the lift coefficient, and C(5.25 ms) denotes the time in milliseconds. Given that the BF collection capacity will be impacted, taking these actions into account may offer a workable approach to identify the ideal solution. SSA has created a monitoring plan for this season as a result. We have a criterion that can

be used to justify utilizing the regional optimal solution without any limitations. The seasonal constant (Sc) is calculated using the formula (51)shown below.

$$S_c^t = \sqrt{\sum_{k=1}^{d}(FS_{at,k}^t - FS_{ht,k})\text{^}2} \tag{51}$$

t = 1, 2, 3, etc. After that, the search is carried out until the SctSmin condition is met. where, according to Eq. (52), Smin is the seasonal constant's minimal value:

$$S_{\min} = \frac{10E^{-6}}{365^{t/\left(\frac{tm}{2.5}\right)}} \tag{52}$$

where the iteration's maximum and current values are denoted by tm and t, respectively.

The BF will no longer be able to search the forest for the finest winter food sources once seasonal monitoring criteria are satisfied. Instead, it will travel at random. It is believed that bacteria that are unable to find pecan trees but still manage to survive would locate a food supply and proliferate. The equivalent BF transformation is modeled by Eq. (53).

$$BF_{nt}^{new} = BF_{LB} + Levy(n) \times (BF_{UB} - BF_{LB}) \tag{53}$$

where the Lévy distribution ensures a more effective and thorough investigation of the searched region.

Step 7: If the condition for halting the algorithm after the maximum number of rounds is not met, steps 4–6 are repeated and the program keeps searching for the best answer. Here is the suggested BFO-HSA CHS method's pseudocode.

Pseudocode of Hybrid Bacteria Foraging Harmony Search

Table 1
Simulation parameter.

Parameters	Values
Total clusters	6
Number of nodes	500
Deployment area	500 × 500
Initial energy	0.1 J
Packet sending rate	1 packet/s
Packet size	512 bytes
Node distribution	Random

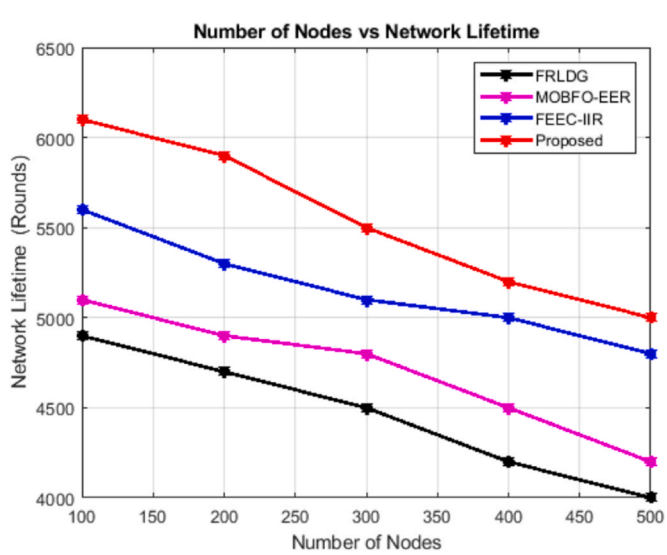

Fig. 5. Number of nodes vs network lifetime.

Optimization Algorithm:

Set up the nodes in the WSN model.
Set the parameter for the algorithm for bacteria foraging.
rmax = Number of cluster head selection iterations
NI – Number of bacterial foraging iterations
Bacterial foraging matrix size is NSq, and BHM is
NSq
for $t \leftarrow 1$ to *rmax* **do**
making a random selection of typical nodes
for $i \leftarrow 1$ to *NI* **do**
for $j \leftarrow 1$ to *NSq* **do**
$FS(j,:) \leftarrow$ a cluster head that was chosen at random for bacteria foraging.
$fObj(j) \leftarrow$ Values of fitness for BF(j,:). Making use of equation (4)
End
Sort the BF rows that correspond to the fObj object and then declare BFht and BFat based on the fitness values.
Declare BFht and BFat depending on the fitness values after sorting the BF rows that correspond to the fObj object.
Pick three sets at random using the Bacteria Foraging method.
Equations (10) and (12) are used to update the BF.
End
Updating the matrix BF with Harmony search methods
choosing the finest cluster heads for data transmission from BF
End

Result and discussion

In WSN, there are numerous SNs, some of which are still active. By routinely observing nodes and categorizing them according to distance, an effective clustering technique reveals the range between nodes.Using the k-medoid method, the cluster's surrounding nodes are gathered, and the CH is chosen based on the SN's energy efficiency. To select the ideal CH at the cluster nodes, the BFO-HSA technique is used. The SN with the highest energy efficiency is decided to be the CH.

The suggested CORP-based BFO-HSA improves the efficiency and reliability of unreliable network communication channels. This routing technique increases the transmission power of other SNs in order to promote network dependability and other communications. The data transmission delay of the synchronizer shortens the communication time between the source SN and the destination node. Traditional cluster

formation and routing methods are outperformed by the CORP and K-medoid algorithms. Traditional data transmission techniques are hampered by high packet loss, transmission latency, and energy instability.(Table 1).

Energy consumption

During CH selection, K-medoid with CORP-based BFO-HSA should choose the best CH based on energy. We therefore employ the CORP-based BFO-HSA approach to determine the shortest path in order to maximize energy consumption. This strategy is not compatible with other routing techniques. Energy use extends the lifespan of wireless sensor networks.

$$E_{con} = (E_{rx} \times number \quad of \quad nodes) + E_{tx} \tag{54}$$

Fig. 4 provides an energy consumption perspective on the intended task and contemporary technologies. Less energy will be used by the 100 CORP-based BFO-HSA nodes that are planned. This technique uses 9.75 mJ as opposed to the 46, 56, and 66 mJ needed by FEECIR, MOBFO-EER, and FRLDG, respectively. While FEEC-IIR, MOBFO-EER, and FRLDG produce 71, 81, and 106 mJ respectively, the suggested method employs 200 nodes and only yields 10 mJ. While the FEEC-IIR, MOBFO-EER, and FRLDG methods each require 99, 109, and 142 mJ, respectively, this method only needs 30.35 mJ at 300 nodes. The approach suggested in this work uses 400 nodes and only requires 37.68 mJ as opposed to the normal versions of FEEC-IIR, MOBFO-EER, and FRLDG, which each require 112, 142, and 160 mJ, respectively. The proposed approach uses 38 mJ at 500 nodes as opposed to the standard FEEC-IIR's 147, 167, and 185 mJ, MOBFO-EER's 167, and FRLDG's 185 mJ. Therefore, the BFO-HSA system based on CORP that is being considered is a very effective way to transmit data.

Network lifetime

The network lifespan in this procedure measures how long the network will last. As the initial sensor stops working, coverage decreases, the network collapses, and the model no longer works in the Eq. (55), the signal-to-noise ratio decreases,

$$E\left[L_{lifetime}\right] = \frac{\sigma_0 - E_{estimate}[E_{RE}]}{CP + \vartheta E[E_{CE}]} \tag{55}$$

Eq. (55) above designates the network's constant infinite power consumption as CP. A popular sensor notification rate called Eestimate [ERE] basically corresponds to energy that isn't used for processing. It displays the volume of data amassed over the same time frame, together with the energy expended. All SNs' E[CE] values correspond to sets of data that were chosen at random. The total amount of inactive energy is denoted by the number 0. In comparison to conventional methods, The network lifetime of the proposed CORP approach is longer. Fig. 5 compares the energy usage of the suggested work with that of existing approaches and shows the network longevity. When data packets are distributed across a wireless sensor network, they must travel further and expend more energy. Less CH energy is used in both CORP-based shortest path discovery and optimal CH selection. Additionally, it increases network lifespan and lowers energy use. Data transfer rates keep rising as networks get older. With lives of 4900, 5100, and 5600 rounds for the most recent FRLDG, MOBFO-EER, and FEEC-IIR, respectively, and a lifetime of 6100 (rounds) for 100-node technology, correspondingly.

Throughput

The throughput is based on the volume of data moving at top speed via the network. It is the amount of time required to send a packet from the sender to the destination, as calculated by Eq. (56).

$$Throughput = \frac{Amount \quad of \quad packet \quad send}{time \quad taken \quad to \quad transmit \quad the \quad packet} \tag{56}$$

In comparison to FEEC-IIR and other approaches, the suggested solution provides a higher throughput on 100 nodes (0.98 Mbps). BFO-HSA with the proper fitness function of CORP serves as the foundation for CH selection. This clustering technique performs excellently. The amount of packets received is significantly impacted by the WSN's lifetime and remaining energy. The throughput diminishes as the number of nodes rises. The effectiveness of the implementation is compared to other solutions in Fig. 6.

Durational delay

Because of transferability and how long it takes to find the shortest path, the CORP technique is not particularly effective. To reduce network redundancy, BFO-HSA, based on CORP, was created. The latency, which is expressed in seconds, indicates the average time it takes for data to travel from source to destination. Eq. (55) uses the time saved during data transmission over the network from the source node to the destination node to get the average delay.

$$Average \quad packet \quad delay = Packet \quad received \quad time - packet \quad send \quad time \tag{57}$$

Fig. 7 compares the CORP-based BFO-HSA that is being suggested and the current techniques in terms of end-to-end latency. The end-to-end latency is reduced by 100 times (0.019 s) using the proposed CORP approach. The figure compares nodes with other conventional techniques. A WSN's time delay increases with the number of nodes. The 100 node latency for contemporary technologies like FRLDG, MOBFO-EER, and FEEC-IIR is 4.17491 s, 2.87848 s, and 2.22530 s, respectively. The proposed CORP-based BFO-HSA reduces packet delays and improves communication line dependability by determining the shortest path.

Packet delivery ratio (PDR)

The suggested cluster-based routing method uses the leftover energy to run huge channels while choosing the best and shortest routing routes to boost packet forwarding's efficiency and reliability. The simulation findings show that adding nodes causes the PDR to decrease. In the formula (58), the outcome is reported as a percentage when the sender's supply of packets is divided by the receiver's receipt of packets.

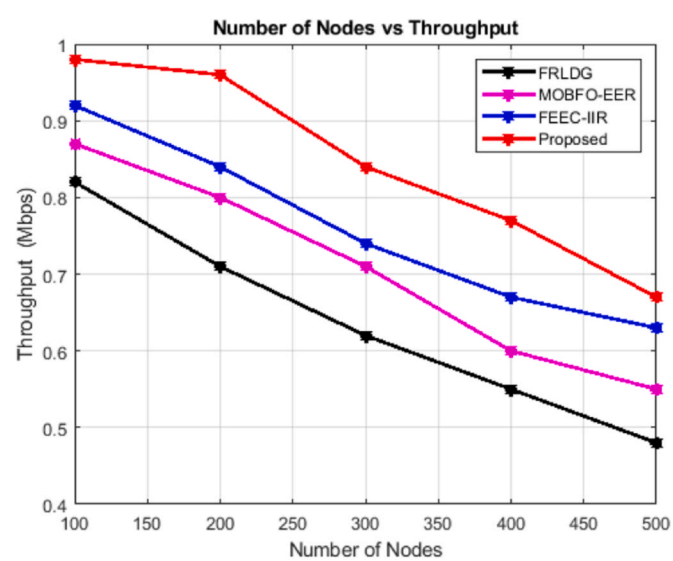

Fig. 6. Number of nodes vs throughput.

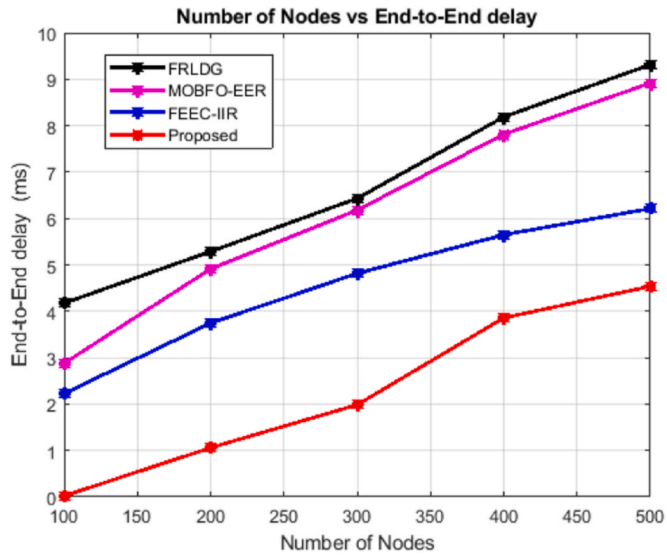

Fig. 7. Number of nodes vs end-to-end delay.

$$PDR = \frac{sum \quad of \quad data \quad packets \quad received \quad by \quad receipt}{sum \quad of \quad data \quad packets \quad transmitted \quad by \quad dispatcher} \tag{58}$$

Fig. 8 shows the results for transmission rate and packet loss rate. The suggested technique utilizes 100 nodes and produces a PDR of up to 98.5 % when compared to currently available energy-efficient devices. Comparing the current procedure to earlier ones, it is more effective. The current FRLDG, MOBFO-EER, and FEEC-IIR have packet forwarding rates of 96 %, 97 %, and 98 % for networks with 100 nodes, respectively. To allow for data flow, the PDR must be high. A high PDR value enables complete target node recovery without data loss. The PDR dramatically rises as node density increases. As a result, the proposed methodology delivers more performance and efficiency when compared to standard approaches.

Packet loss ratio

The amount of data packets lost during transmission between the source node and sink may be measured. Fig. 9 compares the CORP-based BFO-HSA method's packet loss rate to that of the current technology. In comparison to conventional systems, the suggested CORP-based BFO-HSA has a reduced packet loss rate (1.5 %) and is a more effective data transmission system. possess 100 nodes. As there are more nodes, the packet loss rate rises. Packet loss rates for popular algorithms like FRLDG, MOBFO-EER, and FEEC-IIR are 4 %, 3 %, and 2 %, respectively. By comparing the suggested model to conventional approaches, the figure demonstrates a notable reduction in packet loss.

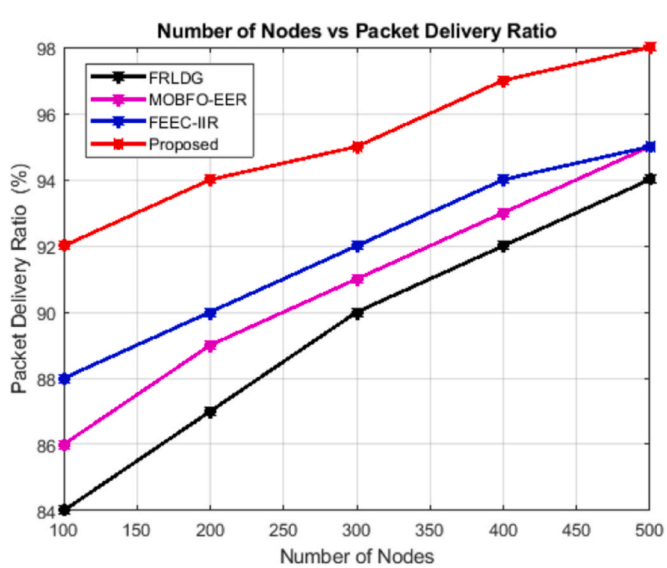

Fig. 8. Number of nodes vs packet delivery ratio.

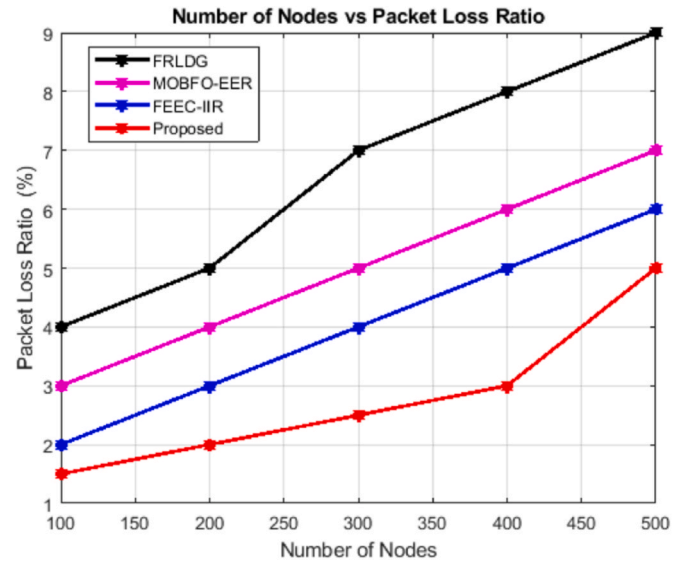

Fig. 9. Number of nodes vs packet loss ratio.

Conclusion

Distributed wireless sensor networks have significant problems in terms of energy efficiency and security. We build clustering, path search, and path maintenance utilising ERFN-CSSBO technology to do this. The quickest path must be established in order to assure data correctness, according the anticipated CORP. Pick the optimal channel grouping and selection strategy to avoid data loss during data transmission from transmitter to receiver. Both k-medoid and BFO-HSA are used at this point. The following performance metrics are used to demonstrate usage: packet delivery rate (98.5 %), packet delay (1.5 s), throughput (1.0 Mbps), energy consumption (30.35 mJ), network lifespan (6100 rounds), and packet loss rate (1.5 %). Do it. a QoS configuration's outcome. This approach is 98.47 % accurate. Therefore, in terms of overall performance, the novel BFO-HSA strategy based on CORP beats the established FRLDG, MOBFO-EER, and FEEC-IIR strategies. Using the hybrid PISA method and adaptive sunflower optimization technology, data noise may be efficiently eliminated. By lowering the energy needed for data transmission, the network's service life is increased. In comparison to LEACH, Hybrid BFO offers nodes with the highest percentage of time that they are 100 % alive as well as long-lived nodes. The CORP-based BFO-HSA strategy beats current methods in terms of WSN performance, according to experimental findings. By extending the network life cycle, upcoming cross-layer based opportunistic routing technologies can enhance WSN performance.

To assure energy-efficient data transfer, the suggested method may be expanded to new proof-free signature techniques with optimum path research. To assure data transmission, computational overhead measurements are also taken into account.

Author's contribution

Architectural design and technical and intellectual content were both contributed by **V.P.R. M.D.** provided direction and advice for the paper's writing.

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Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Since no datasets were created or analyzed for this topic, data sharing is not applicable.

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